Baunah (Bayyinah) IDS / Forensic Dashboard

Purpose

Arabic-first web app to upload evidence CSVs, run XGBoost classification, optionally enrich with Google Gemini, and show MITRE ATT&CK (ICS-style) attack sequence, custody chain, suspicious findings, recommendations, Markdown report, and a visual summary panel.

Technologies - Frontend

React 19, Vite 8, Framer Motion, Lucide React, react-markdown + remark-gfm (Gemini and report rendering). RTL layout and dark UI.

Technologies - Backend

Python, FastAPI, Uvicorn, pandas, XGBoost, scikit-learn, joblib, requests, python-multipart, python-dotenv. Optional Gemini via API key.

Step 1 - Evidence upload

User selects or drags a file. Browser POSTs multipart form to /analyze.

Step 2 - CSV processing

pandas loads CSV, handles missing values, aligns columns to feature_columns.json.

Step 3 - ML inference

RobustScaler + XGBoost model predict probabilities; summary: malicious rate, counts, high-risk flows, inference_mode.

Step 4 - Timeline

Timeline from malicious rows and/or explicit MITRE columns in CSV when present.

Step 5 - MITRE map

Stages normalized: ICS tactic order, one technique per stage, dedupe, Arabic/English labels; optional Gemini narrative and _meta fingerprint.

Step 6 - Gemini (optional)

If GEMINI_API_KEY set: enrich narrative, mitre_attack_map, suspicious_findings, recommendations, final_report, markdown_report; server merges with heuristics.

Step 7 - Custody chain

Server builds chain with UTC timestamps; client fallback if field missing.

Step 8 - Suspicious findings

Cards deduplicated; evidence bullets cleaned from redundant metrics.

Step 9 - Recommendations

Dynamic rules + Gemini; context from techniques and file patterns.

Step 10 - Report

final_report and markdown_report; UI renders Markdown; download as .md file.

Step 11 - Chat

POST /chat-recommendations with session context; assistant reply as Markdown.

Step 12 - Visual panel

Sidebar button opens overlay: KPI ring, file stats, MITRE stage count, timeline excerpt.

Setup - Backend

pip install -r backend/requirements.txt; export GEMINI_API_KEY; uvicorn backend.app:app --reload --port 8000

Setup - Frontend

npm install; npm run dev; optional VITE_IDS_API_URL=http://localhost:8000

Model files

Place under backend/model/: best_xgboost_model.json, robust_scaler.joblib, feature_columns.json (see backend/EXPORT_FROM_NOTEBOOK.md).

Generated for Baunah project. Full detail: README.md in repository root.